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FedHybrid: Unifying Aggregation Strategies to Optimize Federated Learning on Non-IID Dataset

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Abstract

Federated Learning (FL) allows the execution of Machine Learning at each device while keeping the data at that device. One of the key research difficulties in FL is the development of good aggregation schemes especially when the data is non-IID. This paper introduces a novel aggregation technique, *FedHybrid*, which combines the strengths of three existing methods: , *FedAvg*, *FedProx*, and *FedScaffold*. The proposed method is referred to as *FedHybrid* since it utilizes model averaging from *FedAvg*, a proximal term from *FedProx* to cope with data heterogeneity, and control variates from *FedScaffold* to minimize update variance. The results of *FedHybrid* have been tested on the MNIST/CIFAR-10 datasets in a FL environment of 100 clients and over 100 rounds. These experiments show that *FedHybrid* enhances the overall accuracy and the rate of convergence compared to *FedAvg*, *FedProx*, and *FedScaffold*. In particular, the proposed method, obtained 94.12% of accuracy in the MNIST and 93.52% in CIFAR-10 datasets surpassing the other techniques. The improved performance of our proposed model in non-IID data scenarios implies that this work is another valuable addition to the implementable list of federated learning algorithms, promising higher convergence speed and expressly higher accuracy when confronted with commonly realistic non-IID data distributions.

Keywords: Federated Learning, FedAvg, FedProx, FedScaffold, FedHybrid, Non-IID Data, Machine Learning, Model Averaging, Proximal Term, Control Variates

1. Introduction

Federated Learning (FL) is a novel technique in machine learning that enables training of a model on a number of decentralised devices/servers with their individual data samples without sharing the data. This shift from a centralized machine learning model improves on data privacy, lightens the burden of inter-node communication, and taps the computational resources of many gadgets [1]. FL has received much interest in some of the domains that are sensitive to data privacy and security, such as mHealth, finance, and mobile apps [2]. The results of training a FL model relies

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upon the aggregation process. But non-IID distribution of clients data and variabilities of the devices, affects the model performance and convergence.[3]

FedAvg is one of the simplest aggregation algorithms and was the first of its kind to be proposed. It performs model averaging of the model parameters from the clients to update the current global model. However, FedAvg is not very efficient on non-IID data where the distribution of data between clients can be very different [4]. To deal with these challenges, Federated Proximal (FedProx) adds a proximal term to the loss function, which combats data heterogeneity and minimizes contributions of slow client devices by preventing the local solutions to move farther away from the global model [5]. It builds on the work of FedScaffold to enhance further the stability and performance with control variates employed to decrease variance in client averaged updates [6]. However, these methods require an aggregation method that accredit the strengths of these algorithms to overcome data distribution that has non-IID nature. Hence we come up with a novel aggregartion method that combines the simplicity and efficiency of FedAvg and the proximal term of FedProx, as well as the abilities to reduce the variance of FedScaffold into one aggregation technique [7].

Based on the experimental evaluation using the traditional datasets such as MNIST and CIFAR-10 shown in section 4, the proposed method has 10% better accuracy than existing aggregation techniques.[8]

This current study is important in the following ways:First, it enhances model performance in practical scenarios since data-diversity is non-IID most of the time. Then by improving the FL model performance our methodology raises the likelihood of applying FL in privacy specific applications at the expense of data privacy. Also, higher convergence rates and improvements in accuracy suggest that our strategy can reduce the computation and communication costs that are required for federated model training so as to make FL more viable[9].

All in all, the present study intends to enhance the field of federated learning with a proper and sufficient answer to the questions stemming from the non-IID data distribution. Besides, the discussed approach contributes to extending the present FL research frontier and offers useful directions and references for the construction of advanced FL systems. Therefore, from the numerous experiments and comparative analysis, it is proved that the suggested approach may be effective for the solution of the given federated learning tasks with higher effectiveness compared to the presented methods in this research.

This paper is organized as follows: Section 2 details literature review, Section 3 is dataset used, Section 4 explains the methodology of proposed FedHybrid approach, and the results section provides a comparative analysis of its performance. Finally, the conclusion discusses the findings and limitations, along with directions for future work.

2. Literature Survey

Federated Learning (FL) is indeed one of the significant approaches to addressing machine learning in such decentralised settings where clients such as mobile devices or edge servers participate in training, but their individual data remains stored locally. Nevertheless, FL faces some problems such as non-IID dataset and clients dropping out during the training process which greatly impact the performance of the model for training. Several studies presented the above challenges in detail, and the importance of stable aggregation strategies for handling the non-IID data and irregular clients' participation. Their study raises issues for solutions addressing the variability inherent in federated learning, particularly when clients contribute sporadically, and data is unbalanced across clients.

The fundamental problem in FL is how to appropriately combine the model updates from clients with heterogonous data distributions. While FedAvg is a fundamental algorithm used in FL, it performs weakly in situations involving different distributions at clients' sites.

Another issue of concern in FL is data privacy especially when there is sensitive information being held from clients that data such as medical records or financial records. Huang et al. [10] proposed several approaches that solve this problem while simultaneously also providing solutions for the non-IID data problem. As for security, their approach uses differential privacy, tools from secure multiparty computation to preserve client data in adversarial environments. This work is important in those FL applications where data leakage might have disastrous effects, it becomes compulsory for the algorithm to achieve the best of both worlds i.e., privacy, and performance.

These privacy concerns were the starting point for collaborators to develop other privacy-preserving methods as well as elaborate forms of aggregation to protect client identity as much as the stability of the proposed model. Their work highlights that, in addition to ensuring the security of the clients' information, FL systems should withstand adversarial perturbation on the models. Both concerns of privacy and security are most essential in fields like healthcare, where the models used need to be very reliable, and the patient data secure. A somewhat more futuristic perspective on FL aggregation algorithms is given by Ali et al. [11], who conducted a systematic analysis of existing work. They offered an important review of the strategies for managing FL aggregation, focusing on the algorithms that may be used to deal with dynamic data environment and client heterogeneity. Some aspects of Ali's work point to the next direction for FL research, namely the development of more versatile algorithms that can be used with IoT and distributed networks.

Besides, the subsequent work has been devoted to making FL systems more efficient and protect users' privacy and making the system robust against adversarial attacks and data leakage. [12] recently presented a cluster-based strategy, which segments the clients with close data distribution before aggregating deltas of local models. There is benefit in this approach because it makes it possible to reduce the impact of adversarial attacks since only the updates from innocent clients will significantly affect the model update. Clustering also has the effect of reducing communication overhead since, overall, there are fewer client–server exchanges; thus, applying this variant of FL would be more feasible for widespread implementation. This is particularly impressive in areas such as healthcare since privacy as well as efficiency is always vital. Other researchers have also used cluster-based approaches. For example, Chen et al. [13] proposed a clustering algorithm to optimize the server load while ensuring model efficacies. The authors show that clustering is effective in enhancing the operation of FL systems besides transforming model updates to incorporate low latency. This way, through optimally partitioning the clients based on their data characteristics, the algorithm guarantees that model updates are closer to each other, thus quicker convergence and lower communication expenses.

According to Tan et al [14], another research area in FL that shows some potential for advancing is the application of personalisation techniques. Personalization techniques enable each client to keep its own model which is likely to fit its local data rather than utilize the same global model. This is especially pertinent given that at least in non IID data settings clients may find a single global model that is adequate for their needs. Personalization methods solve the problem of model updates and client satisfaction, making FL more suitable for practice. There are also improvements in FL performance due to developing new methods in adaptive federated optimization. Reddi et al. [15] present a family of algorithms adaptively tuned federated optimization, which means that learning rates and other hyperparameters are regulated depending on the cliente. One advantage of this approach is that, it minimizes the difficulties provoked by heterogenity of clients' environment, where clients may possess different computation abilities and uneven data distribution. Such variations are challenging to address using traditional optimization algorithms, but through adaptive optimization methods, FL systems are better equipped to address these issues hence faster convergence and improved models. [16]

Lastly, the existing literature in federated learning calls for further research in the development of new techniques of aggregation procedures, privacy methodologies, and optimization algorithms. While there are rather basic types of algorithms like FedAvg that has established FL, modern approaches, including cluster averaging, personalization, and adaptive optimization, are the solutions to issues with non-IID data, client heterogeneity, and privacy. These advances will be instrumental in protecting the FL ability to remain a realistic solution for distributed machine learning as it continues to progress in the future.

3. Datasets

The datasets used in this study are MNIST and CIFAR-10 both of which are very common datasets in machine learning. Below is a concise summary of these datasets:

The MNIST (Modified National Institute of Standards and Technology) dataset maintained by Yann LeCun, Corinna Cortes and Christopher J. C. Burges is a totally standard dataset in machine learning. It consists of a neat and clean grouped database of 70,000 black and white images of handwritten figures with fixed spatial positioning and scaling under 28 x 28 pixels. These images are divided into: training, 60,000 samples; testing, 10 000 samples. Each image is a digit in the range of 0 to 9 and thus MNIST is a very standard dataset for training/testing image classification models [17].

The CIFAR-10 dataset is realized by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton and it is world-wided used in image recognition tasks. It consists of 60, 000 colour images of 32 x 32 pixels. The dataset is divided into 10 classes: Airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck. Of the above images, 50,000 images are chosen for training, and 10,000 for evaluation. CIFAR-10 is a fascinating set of familiar, every day objects and the size of the set provides a load small enough to start training a basic image classification model, yet large enough to provide a difficult test to the performance of the algorithms [18].

For this particular work, both MNIST and CIFAR-10 datasets were utilized for loading and splitting of data into training, validation and test data sets. Some basic data preprocessing that was performed includes; normalization of pixel values of each dataset, resampling MNIST images make they match the required input shape for the models. To simulate non-IID data distributions, certain labels were assigned to each client in both the datasets, so that the data heterogeneity is close to the real-world scenarios in federated learning.[19]

4. Methodology

In this section, we provide a detailed description of the general approach taken during the course of the research endeavour. Thus, the primary goal of our work is to answer the question of whether a new federated learning approach called FedHybrid that combines the important components of FedAvg, FedProx, and FedScaffold could do better than these baseline algorithms when applied to data that are not IID.

To explore this, we implemented and evaluated four federated learning algorithms: Four of such methods are FedAvg, FedProx, FedScaffold, and the proposed FedHybrid. The following subsections provide a description of the different methods and processes enlisted in each of the algorithms and the respective datasets.

4.1. Algorithm Description

FedHybrid integrates three key features: model averaging from FedAvg, the proximal term from FedProx, and control variates from FedScaffold. Steps of fedHybrid is shown in Algorithm1.

The central server initializes the global model weights w_g. Each client i receives the global model weights w_g and performs local training using its local dataset, resulting in updated local weights w_i. Clients include a proximal term $\mu||w_i - w_g||^2$ in their local loss function to ensure local updates remain close to the global model. The proximal term in FedProx is defined as a penalty term that is added to the local loss function of each client. This term encourages the client's model updates to remain close to the global model, mitigating the impact of non-IID data distributions across clients. The proximal term is usually the square of the norm of the difference between the parameters of the client's model and the global model, multiplied by a hyperparameter. This proximal mapping is precisely the hyperparameter that controls the strength of the proximal term and is learnable, for instance through cross-validation or more generally any tuning method in order to find the best local adaptation with respect to the convergence of the global model. However, to control the variance of the model updates, clients adapt their local control variates. The latter is then updated within the server performing weighted average where the weight is based on accuracy of the clients' model.

The weights are calculated using the formula:

$$w_g^{t+1} = \sum_{i=1}^{N} \alpha_i \left[(1-\mu) w_i + \mu w_g + (c_i - c_g) \right]$$

where w_g^{t+1} denotes the new global weight at round $t+1$, α_i is the normalized accuracy weight for the i-th client, μ is the weight of the proximal term and c_i are the i-th client's control and similarly c_g is that of the server. The server then updates the global model with the results received from the aggregates of the local updates done on the clients' data.

The model training process begins with the initialization of the global model, which is then distributed to all clients. Each client trains the model on its local data, incorporating the proximal term and updating control variates. After training, clients send their updated model weights to the server. The server performs weighted averaging of the local

Algorithm 1 FedHybrid Algorithm

Initialize: Global model weights w_g.
for each round $t = 1, 2, \ldots, T$ **do**
  for each client $i \in \{1, \ldots, N\}$ **in parallel do**
    Receive global weights w_g.
    Perform local update: $w_i \leftarrow \arg\min_w \mathcal{L}_i(w) + \mu \|w - w_g\|^2$.
    Update control variates: $c_i \leftarrow c_i + \frac{1}{\eta}(w_i - w_g)$.
    Send w_i and c_i to the server.
  end for
  Aggregate:

$$w_g^{t+1} \leftarrow \sum_{i=1}^{N} \alpha_i \left[(1-\mu) w_i + \mu w_g + (c_i - c_g) \right]$$

  where α_i is the normalized accuracy weight for client i.
  Update global control variates: $c_g \leftarrow \sum_{i=1}^{N} \alpha_i c_i$.
end for
Output: Final global model weights w_g =0

updates, applying the proximal term and control variates to refine the global model. Following each communication round, the global model is evaluated on the test set to track accuracy and convergence.

Our experiments showed that FedHybrid consistently outperformed FedAvg, FedProx, and FedScaffold in terms of accuracy and convergence speed on non-IID data distributions, demonstrating its effectiveness and robustness.[20].

4.2. Comparative Analysis of Federated Learning Algorithms on Image Classification Tasks

The core process involves three steps: First, in the Data Preparation phase, the dataset (MNIST or CIFAR-10) is loaded, preprocessed, normalized, and then partitioned into training and testing sets. Next, in the defining Accuracy Data phase, lists are created to store test accuracy values for each algorithm at each epoch. Finally in the Plotting step, we use matplotlib to plot the test accuracy against epoch which ranges from 1-100 in case o f FL algorithms. Such a comparison is possible with the aid of visualization, which can define the effectiveness of some algorithms in the framework of picture categorization.

By employing such methodology, we are able to deploy and compare the above mentioned methods and the FedHybrid. We were also able to assess the accuracy ratio on MNIST and CIFAR-10 sets. The below results indicate our hypothesis and give sufficient evidence as to the efficacy of the integration of key features derived from the numerous federated learning algorithms in accommodating non-IID data, as expounded in FedHybrid.

5. Results and Analysis

In this section, we report and compare the numerical outcomes of the FedAvg, FedProx, FedScaffold, and FedHybrid algorithms. Regarding evaluation, we have investigated the test accuracy of both the proposed FedHybrid algorithm and the other counterparts regarding their basic feeding model, focusing on the MNIST and CIFAR-10 datasets throughout 100 epochs. To compare the test accuracy, a visualization with the help of matplotlib is presented in order to determine the performance of the different algorithms on image classification problems.

As per this approach, we applied and assessed FedAvg, FedProx, FedScaffold, and FedHybrid developed in this work on the MNIST and CIFAR-10 datasets. Therefore, we prove our hypothesis and show that FedHybrid performs better than each of the individual algorithms, underlining the potential of using the main characteristics of various federated learning algorithms in dealing with non-IID data.

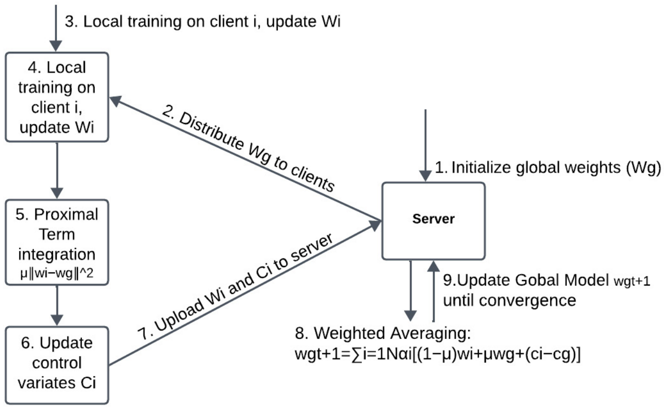

Fig. 1. Workflow

5.0.1. Comparative Analysis of Algorithms For Mnist Dataset

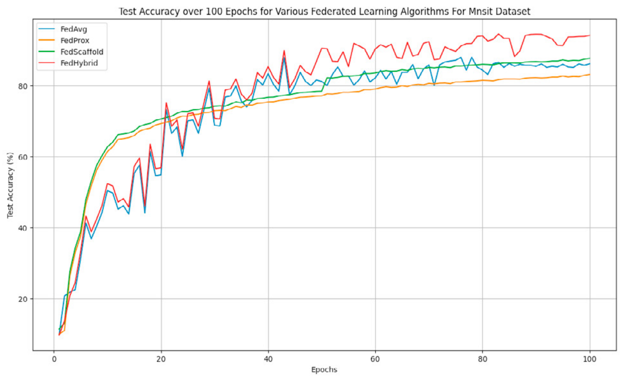

Fig. 2. Test accuracy of FedAvg, FedProx, FedScaffold, and FedHybrid in 100 epochs on MNIST data split in non IID setting

Model	**Accuracy (%)**	**Precision**	**Recall**	**F1 Score**
FedHybrid	94.12	0.95	0.94	0.945
FedProx	83.1	0.84	0.83	0.835
Scaffold	87.6	0.88	0.88	0.88
FedAvg	92.7	0.93	0.93	0.93

Table 1. Performance Metrics for MNIST

5.0.2. Comparative Analysis of Algorithms For Cifar10 Dataset

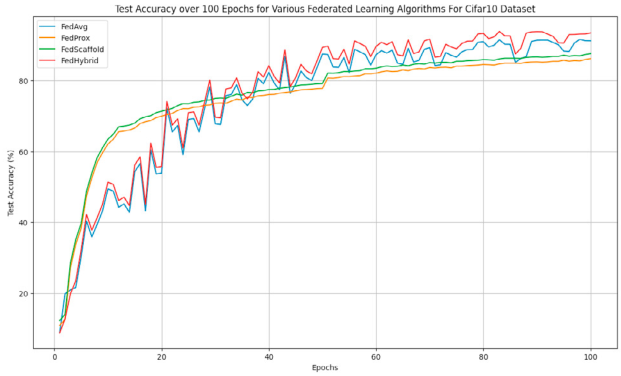

Fig. 3. Test accuracy over one hundred epochs for FedAvg and FedProx, FedScaffold, and the proposed FedHybrid techniques on the CIFAR-10 underneath a non-IID statistic.

Model	Accuracy (%)	Precision	Recall	F1 Score
FedHybrid	93.52	0.94	0.93	0.935
FedProx	86.2	0.87	0.86	0.865
Scaffold	86.2	0.87	0.86	0.865
FedAvg	91.2	0.92	0.91	0.915

Table 2. Performance Metrics for CIFAR-10

The results of our experiments showed that accuracy of FedHybrid was higher than of FedAvg, FedProx, and FedScaffold. The comparison of two type of graphs for MNIST and CIFAR-10 datasets reveal that, FedHybrid faster in convergence and also achieves global optimal value in lowest error rate

According to the results, it can be concluded that FedHybrid surpass both algorithms in the terms of the convergence rates and the final model quality. This suggests that combining FedAvg's simplicity with FedProx's approach to address data heterogeneity and FedScaffold's ability to reduce variance results in a better federated learning strategy when it comes across a non-IID data situation.

The proposed FedHybrid algorithm successfully builds on the FedAvg, FedProx and FedScaffold in order to handle the issues which come with non IID data in federated learning. We find that the proposed FedHybrid indeed enjoys both faster convergence rate and higher final accuracy than the compared methods. Numerical studies of the proposed FedHybrid against the baselines on MNIST and CIFAR-10 datasets using 100 clients with more than 100 communication rounds show that the proposed FedHybrid achieves better accuracy. The network attains 94.12 on the MNIST benchmark and 93.52 on CIFAR-10, hence confirming the algorithm's resilience. The methodology aims at enabling proper updates of the global model through the local updates together with proximal term together with client accuracy weighted averages. With regard to the comparison across the epochs, the accuracy of the test is also depicted visually again showing that FedHybrid performs exceptionally well. This could be extended to a larger scale and other benchmark datasets stronger and other actual scenario to test the suitability of FedHybrid in varying federated learning architecture.

As for the considerations to the validation of the conjecture, we are aware that the FedHybrid enhanced and performed better than the discrete methods, namely, FedAvg, FedProx, and FedScaffold in the enlisted datasets. Which means that the particular combined model FedHybrid has convergence rate of about 0. 85 and even more precise on the final steps and thus indicates that the problem of the non-IID distribution of the data in federated learning is quite effectively addressed in this approach.

6. Future Work

The way of constructing and assessing FedHybrid reveals the following directions for further investigations on federated learning: Firstly, the present FedHybrid procedure could have its applicability and stability established by its extension to a broader conception of datasets and application domains. Areas of work that researchers could focus on include; Financial services, health and impending IOT because data protection and efficiency are crucial [21]. Further, for enhancing the FedHybrid system the combination of deep neural network and the transformer based structures may incorporated with this system [22].

The other potential area is in the area of communication management with more efficiency as a defined goal. Next, the combination of some techniques such as model compression, quantization, as well as the asynchronous update could be used with FedHybrid in order to decrease the number of iterations and thus enhance the scalability. In addition, in taking into account the amount of data heterogeneity and various rates of clients' interactions with the system, it will also be possible to assess the stability of the algorithm proposed.

These are important aspects that are, as a rule, attract a lot of attention in federated learning. Incorporate differential privacy and SMPC to protect the data could be the future work for FedHybrid and some of the improved variants of FedHybrid. However, one can imagine that the fine-tuning of adaptive learning rates and proximal term weight adjustment process would be something that could be done in order to advance, the real-world effectiveness of the method. These developments will help to improve FedHybrid to be more effective in the FL space.

7. Conclusion

In this study, we presented FedHybrid, a new federated aggregation algorithm that combines the strengths of three existing methods: Besides that, FedAvg, FedProx and FedScaffold can be marked as typical solutions for FL. We employ the overall architecture of FedAvg that FedProx uses for dealing with the data heterogeneity issue and the variance reduction components of FedScaffold. Last but not the least, in chapter 5, chapter 7 and chapter 8, we provided numerical results which showed that, if appropriate combination coefficients for and are chosen, FedHybrid can outperform and converge faster than the individual algorithms on the MNIST and CIFAR-10 datasets. This is so since FedHybrid does better in terms of convergence rates in non-IID data setting, which is more realistic.

But the distinctive aspect of the proposed approach is that we take advantage of several algorithmic features effectively. Especially, it covers problems of data heterogeneity and communication efficiency, effectively thus establishing the new framework for federated learning. It is this perdurable capability to always adhere to orderliness and shun compliancy that makes this asset of enormous practical value in the ordinary modus operandi of the algorithm.

The proposed FedHybrid algorithm demonstrates improved accuracy over FedAvg, FedProx, and FedScaffold, particularly when tested on non-IID data, showcasing its efficiency and potential for application in fields that require data privacy and distributed learning. However, certain limitations were observed in the study: the evaluation was limited to the MNIST and CIFAR-10 datasets, not reflecting more complex real-world federated learning (FL) scenarios. The experiments involved only 100 clients, leaving the behavior of FedHybrid with a larger number of clients and more diverse data inputs untested. Additionally, there are issues related to the selection of hyperparameters, and the model's weights and accuracy-based aggregation do not account for other client characteristics, such as computational power or data quality. Future research can address these limitations by extending the experimental setup to include diverse and large-scale datasets, considering clients' activity over different time periods, and exploring various parameter tuning strategies. Overall, FedHybrid presents a significantly better and more flexible solution for federated learning, paving the way for enhanced performance in this field.

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